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FAQ: How to operate AX5043 at 27 MHz - (custom matching networks for AX5043)

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Title
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Answer

Question

What is the procedure for operating AX5043 at 27 MHz?

Recommendation

27 MHz is at the low end of the frequency range that AX5043 was designed for. Because applications are seldom built for this low range, ON Semiconductor has not provided a formal reference design. However, some work has been done to configure the AX5043 at this frequency. This post summarizes what work has been done, and the limitations of operating at this frequency.

The first important note to make is that, as can be seen in the chart below (which is included in the AX5043 datasheet), lower frequencies require larger external tuning inductors for the internal VCO. Additionally, the gain of the VCO (K_VCO) decreases dramatically as the inductance size increases. This means that the PLL has a much smaller range of available frequencies to lock to for a given inductor value. One issue we've seen is that unless extremely tight tolerances are used on this tuning inductor, a test bench design at 27 MHz may experience issues in production. This is because the slight variations in inductance place the entire PLL tuning range outside of the desired RF operating frequency band.

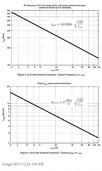

Figure 1. RF Gain and Resonance Results, Tuned Frequency vs. L_tune